

CBSE Practice Worksheet

Class 8 | Science | Difficulty: Hard | Set D

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. Which principle explains the working of a hydraulic lift?
(A) Archimedes
(B) Pascal
(C) Bernoulli
(D) Newton
2. Which quantum number determines the shape of an orbital?
(A) Principal
(B) Azimuthal
(C) Magnetic
(D) Spin
3. Which organelle is involved in programmed cell death?
(A) Lysosome
(B) Mitochondria
(C) Peroxisome
(D) Ribosome
4. What is the wavelength of Lyman series limit?
(A) 91.2 nm
(B) 121.6 nm
(C) 656.3 nm
(D) 364.6 nm
5. What is the magnetic quantum number for 3d orbital?
(A) -2 to +2
(B) -3 to +3
(C) -1 to +1
(D) 0 only
6. What is the efficiency of an ideal Carnot engine operating between 300K and 400K?
(A) 25%
(B) 33%
(C) 50%
(D) 75%
7. Which enzyme is responsible for DNA replication?
(A) DNA ligase
(B) DNA polymerase
(C) Helicase
(D) Primase
8. What is the wavelength of de Broglie wave for an electron?
(A) $\lambda = h/mv$
(B) $\lambda = mv/h$
(C) $\lambda = hv/m$

(D) $\lambda = m/h\nu$

9. Which law relates induced EMF to rate of change of magnetic flux?

- (A) Lenz's law
- (B) Faraday's law
- (C) Ampere's law
- (D) Ohm's law

10. What is the oxidation state of Mn in KMnO_4 ?

- (A) +2
- (B) +4
- (C) +6
- (D) +7

Section B: Fill in the Blanks (1 mark each)

- 11. The IUPAC name of $\text{CH}_3\text{CH}_2\text{CHO}$ is ____.
- 12. Raoult's law applies to ____ solutions.
- 13. The magnetic field inside a solenoid is given by $B = \mu \blacksquare$ ____.
- 14. The Krebs cycle occurs in the ____ of mitochondria.
- 15. The band gap in semiconductors is typically ____ eV.
- 16. The enzyme that converts prothrombin to thrombin is ____.
- 17. For a reversible process, entropy change of the universe is ____.
- 18. The process of splitting water during photosynthesis is called ____.

Section C: Short Answer Questions (2 marks each)

- 19. Describe the light-dependent and light-independent reactions of photosynthesis.
- 20. Explain the mechanism of DNA replication.
- 21. Describe the mechanism of nerve impulse transmission.
- 22. Describe electromagnetic induction and derive Faraday's law.
- 23. Explain hybridization with examples of sp , sp^2 , and sp^3 .
- 24. Derive the expression for electric field due to a dipole.
- 25. Derive the lens formula and explain its applications.

--- End of Worksheet ---

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